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Individual Assessment Coversheet

To be attached to the front of the assessment.

Campus: EEDUVOS ONLINE

Faculty: Infomation Technology

Module Code: ITPPA1-44

Group: Group 2

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Indicate	Yes	No
Plagiarism report attached	X	

Declaration:

I declare that this assessment is my own original work except for source material explicitly acknowledged. I also declare that this assessment or any other of my original work related to it has not been previously, or is not being simultaneously, submitted for this or any other course. I am aware of the AI policy and acknowledge that I have not used any AI technology to generate or manipulate data, other than as permitted by the assessment instructions. I also declare that I am aware of the Institution's policy and regulations on honesty in academic work as set out in the Conditions of Enrolment, and of the disciplinary guidelines applicable to breaches of such policy and regulations.

Signature	Date
Sachin Jody Abrahams	02/11/25

Lecturer's Comments:

Marks Awarded:	%

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Question 1:

Question1.cpp

C: > Users > sachi > OneDrive > Visual Studio Project Sachin Abrahams EDUV6554640 > Question1.cpp > ...

```cpp
#include <iostream>
#include <iomanip>
#include <string>

using namespace std;

const int MAX_ITEMS = 10;
const int CATALOG_SIZE = 8;
const double DISCOUNT_THRESHOLD = 2000.0;
const double DISCOUNT_RATE = 0.15;

// Gadget names and prices
string gadgetNames[CATALOG_SIZE] = {
    "Wireless Mouse", "Bluetooth Speaker", "USB Flash Drive", "Gaming Keyboard",
    "Noise-Cancelling Headphones", "Smartwatch", "External Hard Drive", "Portable Charger"
};

double gadgetPrices[CATALOG_SIZE] = {
    150.0, 250.0, 100.0, 500.0, 750.0, 1200.0, 1000.0, 300.0
};

void displayCatalogue() {
    cout << "\nGadget Catalogue:\n";
    for (int i = 0; i < CATALOG_SIZE; ++i) {
        cout << "  " << (i + 1) << ". " << gadgetNames[i]
             << " - R" << fixed << setprecision(2) << gadgetPrices[i] << endl;
    }
}

int main() {
    string firstName, lastName;
    int itemNumbers[MAX_ITEMS], quantities[MAX_ITEMS];
    double subtotal = 0.0;

    // Prompt user details
    cout << "Enter your first name: ";
    cin >> firstName;
    cout << "Enter your last name: ";
    cin >> lastName;

```

```
    // Display catalogue
    displayCatalogue();

    cout << "\n Select up to 10 items by entering item number and quantity.\n";

    for (int i = 0; i < MAX_ITEMS; ++i) {
        int itemNum, qty;

        cout << "\nItem " << (i + 1) << " - Enter item number (1-8, or 0 to stop): ";
        cin >> itemNum;

        if (itemNum == 0) break;

        // Validate item number
        while (itemNum < 1 || itemNum > CATALOG_SIZE) {
            cout << " Invalid item number. Please enter a number between 1 and 8: ";
            cin >> itemNum;
        }

        cout << "Enter quantity: ";
        cin >> qty;

        // Validate quantity
        while (qty < 1) {
            cout << " Quantity must be at least 1. Try again: ";
            cin >> qty;
        }

        itemNumbers[i] = itemNum;
        quantities[i] = qty;

        subtotal += gadgetPrices[itemNum - 1] * qty;
    }

    // Apply discount if applicable
    double discount = 0.0;
    if (subtotal > DISCOUNT_THRESHOLD) {
        discount = subtotal * DISCOUNT_RATE;
    }

    double finalTotal = subtotal - discount;

    // Display results
    cout << "\n Purchase Summary for " << firstName << " " << lastName << ":\n";
    cout << "----------------------------------------\n";
    cout << "Subtotal:         R" << fixed << setprecision(2) << subtotal << endl;
    cout << "Discount Applied: R" << discount << endl;
    cout << "Final Total:      R" << finalTotal << endl;
    cout << "----------------------------------------\n";

    return 0;
}
```

```
Enter your first name: sachin
Enter your last name: abrahams

Gadget Catalogue:
  1. Wireless Mouse - R150.00
  2. Bluetooth Speaker - R250.00
  3. USB Flash Drive - R100.00
  4. Gaming Keyboard - R500.00
  5. Noise-Cancelling Headphones - R750.00
  6. Smartwatch - R1200.00
  7. External Hard Drive - R1000.00
  8. Portable Charger - R300.00

 Select up to 10 items by entering item number and quantity.

Item 1 - Enter item number (1-8, or 0 to stop): 2
Enter quantity: 3

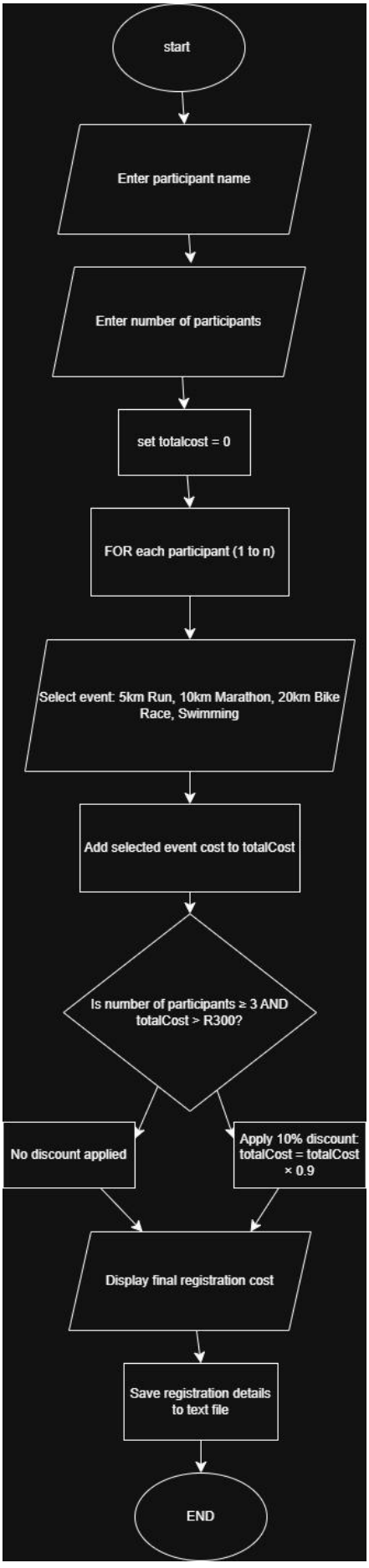
Item 2 - Enter item number (1-8, or 0 to stop): 4
Enter quantity: 5

Item 3 - Enter item number (1-8, or 0 to stop): 0

 Purchase Summary for sachin abrahams:
----------------------------------------
Subtotal:         R3250.00
Discount Applied: R487.50
Final Total:      R2762.50
----------------------------------------
```

Question 2

Flowchart:



Pseudo-code:

1. Start

2. Prompt user to enter their name

3. Prompt user to enter the number of participants

4. Set totalCost = 0

5. FOR each participant from 1 to number of participants DO

   a. Display event options:

      - 5 km Fun Run: R50

      - 10 km Marathon: R100

      - 20 km Bike Race: R150

      - Swimming Competition: R80

   b. Prompt user to select an event

   c. Add selected event cost to totalCost

6. IF number of participants ≥ 3 AND totalCost > R300 THEN

   a. Apply 10% discount:

      totalCost = totalCost × 0.9

7. Display final registration cost on screen

8. Save registration details (name, number of participants, selected events, final cost) to a text file

9. End

Question 3:

```cpp
#include <iostream>      // Includes input/output stream for cin and cout
#include <algorithm>     // Includes functions like max_element and min_element

using namespace std;     // Allows use of standard namespace (e.g., cout, cin)

int main(){

    // Array of strings to label each day of the week
    string days[7] = {"Day1: ", "Day2: ", "Day3: ", "Day4: ", "Day5: ", "Day6: ", "Day7: "};

    // Array to store temperature values for 7 days
    int temps[7] = {};

    // Attempt to calculate array size (incorrectly uses sizeof(int) instead of sizeof(string))
    const int SIZE = sizeof(days)/sizeof(int);

    int sum = 0;         // Variable to store the sum of all temperatures
    int average = 0;     // Variable to store the average temperature
    int entry = 0;       // Unused variable (can be removed)

    // Loop to input temperatures for each day
    for(int i = 0; i < 7; i++) {
        cout << "Please enter the tempreture for the day:" << endl;  // Prompt for temperature
        cout << days[i];                                             // Display current day label
        cin >> temps[i];                                             // Store input in temps array
        sum += temps[i];                                             // Add temperature to running total
    }

    // Calculate average temperature (integer division)
    average = sum / 7;

    // Display average temperature with degree symbol (char(248)) and Celsius unit
    cout << "The average tempreture for the week is: " << average << char(248) << "C" << endl;

    // Display highest temperature using max_element from <algorithm>
    cout << "The maximum for the week is: " << *max_element(temps, temps + 7) << char(248) << "C" << endl;

    // Display lowest temperature using min_element from <algorithm>
    cout << "The minimum for the week is: " << *min_element(temps, temps + 7) << char(248) << "C" << endl;

    // Loop to display all recorded temperatures in the order they were entered
    for(int j = 0; j < 7; j++){
        cout << days[j] << " " << temps[j] << char(248) << "C" << endl;  // Display day and temperature
    }
```

```
Please enter the tempreture for the day:
Day1: 20
Please enter the tempreture for the day:
Day2: 21
Please enter the tempreture for the day:
Day3: 25
Please enter the tempreture for the day:
Day4: 23
Please enter the tempreture for the day:
Day5: 21
Please enter the tempreture for the day:
Day6: 22
Please enter the tempreture for the day:
Day7: 23
The average tempreture for the week is: 22°C
The maximum for the week is: 25°C
The minimum for the week is: 20°C
Day1:  20°C
Day2:  21°C
Day3:  25°C
Day4:  23°C
Day5:  21°C
Day6:  22°C
Day7:  23°C
```

Question 4:

```
// Function to create a new member account
void createAccount(string &name, int &memNumber, int &credits){
    cout << "-- Create Account--" << endl;

    // Prompt user for name
    cout << "Please enter your name: ";
    cin >> name;

    // Prompt user for membership number
    cout << "Please enter your MemberShip Number: ";
    cin >> memNumber;

    // Prompt user for initial session credits
    cout << "Please enter your session credits: ";
    cin >> credits;

    // Validate that credits are non-negative
    if(credits < 0){
        cout << "Invalid session credits added. Must be above (1). \n";
        cout << "Please enter your session credits: ";
        cin >> credits;
    }
}

// Function to add session credits to an existing account
int addSessionCredits(int &credits, int &amountAdded){
    cout << "--Add Session Credits--" << endl;

    // Prompt user for amount to add
    cout << "Please enter the amount of credits you would like to add: ";
    cin >> amountAdded;

    // Add credits to current balance
    credits += amountAdded; // Adds the entered amount to the existing credits

    // Validate that the added amount is non-negative
    if(amountAdded < 0){
        cout << "Please enter a valid session credit amount (>=0).\n";
        cout << "Please enter the amount of credits you would like to add: ";
        cin >> amountAdded;
    }
    return 0;
}
```

```cpp
// Function to book sessions and deduct credits
int bookSession(int &credits, int &sessionsBooked){
    cout << "--Book Session--" << endl;
    cout << "Welcome! Thank you for wanting to book a session.\n";

    // Prompt user for number of sessions to book
    cout << "How many session would you like to book?: ";
    cin >> sessionsBooked;

    // Deduct booked sessions from available credits
    credits -= sessionsBooked;

    // Validate that credits are still non-negative after booking
    if(credits < 0){
        cout << "Invalid credits. Please enter a valid amount of credits available.\n";
        cout << "How many session would you like to book?: ";
        cin >> sessionsBooked;
        credits -= sessionsBooked;
    }
    return 0;
}

// Function to check remaining session credits
void checkCredits(int &credits){
    cout << "--Check Credits--" << endl;

    // Display remaining credits if available
    if(credits > 0){
        cout << "Remaining session credits: " << credits << endl;
    } else {
        // Inform user if no credits are available
        cout << "no credits available.";
    }
}

// Function to display account details
void displayAccountDetails(string &name, int &memNumber, int &credits){
    cout << "--Account Details--" << endl;
    cout << "Member Name: " << name << endl;
    cout << "Membership number: " << memNumber << endl;
    cout << "credits avaliable: " << credits << endl;
}
```

```cpp
int main(){
    // Declare variables used in account management
    string name;
    int memNumber;
    int credits = 1; // Initial default credit
    int amountAdded = 0;
    int sessionsBooked;

    // Declare variables for menu control
    int accountCreate;
    int addCredit;
    int sessionBook;
    int creditCheck;
    int displayDetails;
    string exit;
    int option;

    // Loop to keep showing menu until user chooses to exit
    do{
        // Display main menu
        cout << "--Fitness MemberShip Managemant System--" << '\n';
        cout << "1. Create MemberShip Account." << '\n';
        cout << "2. Add Session Credits" << '\n';
        cout << "3. Book a Session" << '\n';
        cout << "4. Check credits" << '\n';
        cout << "5. Display Account Details." << endl;

        // Ask user whether to exit or continue
        cout << "Press (e) to exit, (c) to continue:";
        cin >> exit;
        cout << endl;

        // If user enters "e", loop will break.
        if(exit == "e"){
            break;
        }

        // Prompt user for menu option
        cout << "Please enter a option: ";
        cin >> option;
```

```
132         // Validate that option is within valid range
133         if(option <=0 || option >5){
134             cout << "Invalid input please enter a number between 1 and 5.";
135             break;
136         };
137 
138         // Execute corresponding function based on user choice
139         switch(option){
140             case 1:
141                 createAccount(name, memNumber, credits);
142                 cout << endl;
143                 break;
144             case 2:
145                 addSessionCredits(credits, amountAdded);
146                 cout << endl;
147                 break;
148             case 3:
149                 bookSession(credits, sessionsBooked);
150                 cout << endl;
151                 break;
152             case 4:
153                 checkCredits(credits);
154                 cout << endl;
155                 break;
156             case 5:
157                 displayAccountDetails(name, memNumber, credits);
158                 cout << endl;
159                 break;
160             default:
161                 // Catch-all for unexpected input
162                 cout << "Invalid input, enter value between 1-5.";
163         }
164 
165     } while(exit != "e"); // Continue loop until user enters 'e' to exit
166 
167     return 0;
168 }
```

```
--Fitness MemberShip Managemant System--
1. Create MemberShip Account.
2. Add Session Credits
3. Book a Session
4. Check credits
5. Display Account Details.
Press (e) to exit, (c) to continue:c

Please enter a option: 1
-- Create Account--
Please enter your name: sachin
Please enter your MemberShip Number: 1234
Please enter your session credits: 3

--Fitness MemberShip Managemant System--
1. Create MemberShip Account.
2. Add Session Credits
3. Book a Session
4. Check credits
5. Display Account Details.
Press (e) to exit, (c) to continue:2

Please enter a option: 2
--Add Session Credits--
Please enter the amount of credits you would like to add: 2

--Fitness MemberShip Managemant System--
1. Create MemberShip Account.
2. Add Session Credits
3. Book a Session
4. Check credits
5. Display Account Details.
Press (e) to exit, (c) to continue:3

Please enter a option: 3
--Book Session--
Welcome! Thank you for wanting to book a session.
How many session would you like to book?: 2

--Fitness MemberShip Managemant System--
1. Create MemberShip Account.
2. Add Session Credits
3. Book a Session
4. Check credits
5. Display Account Details.
Press (e) to exit, (c) to continue:c

Please enter a option: 4
```

```
Please enter a option: 4
--Check Credits--
Remaining session credits: 3

--Fitness MemberShip Managemant System--
1. Create MemberShip Account.
2. Add Session Credits
3. Book a Session
4. Check credits
5. Display Account Details.
Press (e) to exit, (c) to continue:c

Please enter a option: 5
--Account Details--
Member Name: sachin
Membership number: 1234
credits avaliable: 3
```

Refernces:

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